

Generation Interconnection System Impact Study Report

For Queue Project Q11-0041

Hell Creek 138 kV

July 16, 2012

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0041. The Interconnection Customer proposes a 1500.0 MW natural gas combined cycle facility interconnecting at the Hell Creek 138 kV substation in the New Jersey area. The request was received on May 27, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 35 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$93,751,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Hell Creek-Windhub 138.0 kV
Contingency: Hell Ck Substation-Framingham 138.0 kV
Loading: 841.9% (Rating: 175 MVA)
DFAX: 100.0%
Pre-project loading: 17.6%

2. Facility: Hell Crk-Framingham 138.0 kV
Contingency: Hell Creek-Windhub 138.0 kV
Loading: 746.2% (Rating: 197 MVA)
DFAX: 100.0%
Pre-project loading: 15.6%

3. Facility: Framingham-Gaffney 138.0 kV
Contingency: Hell Crk-Windhub 138.0 kV
Loading: 657.5% (Rating: 219 MVA)
DFAX: 100.0%
Pre-project loading: 28.0%

4. Facility: Gaffney-Reno 138.0 kV
Contingency: Hell Creek-Windhub 138.0 kV
Loading: 611.9% (Rating: 231 MVA)
DFAX: 100.0%
Pre-project loading: 36.9%

5. Facility: Centralf-Windhub 138.0 kV
Contingency: Hell Crk-Framingham 138.0 kV Line
Loading: 510.9% (Rating: 186 MVA)
DFAX: 62.8%
Pre-project loading: 4.2%

6. Facility: Reno-Vincent 138.0 kV L
Contingency: Windhub-Hell Creek 138.0 kV
Loading: 350.7% (Rating: 190 MVA)

DFAX: 43.2%

Pre-project loading: 9.5%

7. Facility: Greenwood-Centralf 138.0 kV

Contingency: Hell Crk-Framingham Substation 138.0 kV

Loading: 302.3% (Rating: 162 MVA)

DFAX: 33.0%

Pre-project loading: 3.7%

8. Facility: Reno-Jackson Square 138.0 kV

Contingency: Hell Crk-Windhub 138.0 kV

Loading: 290.0% (Rating: 249 MVA)

DFAX: 56.8%

Pre-project loading: 53.0%

9. Facility: Dillion-Windhub 138.0 kV L

Contingency: Windhub-Centralf 138.0 kV

Loading: 246.4% (Rating: 217 MVA)

DFAX: 38.9%

Pre-project loading: 22.3%

10. Facility: Country Club 2 Tap-Dillion 138.0 kV

Contingency: Windhub-Centralf 138.0 kV

Loading: 217.5% (Rating: 233 MVA)

DFAX: 38.9%

Pre-project loading: 33.2%

11. Facility: Country Club 2 Tap-Jackson Square 138.0 kV

Contingency: Windhub-Centralf 138.0 kV L

Loading: 209.5% (Rating: 237 MVA)

DFAX: 38.9%

Pre-project loading: 37.0%

12. Facility: Centralf-Universit Wps 138.0 kV

Contingency: Centralf-Greenwood 138.0 kV

Loading: 208.5% (Rating: 202 MVA)

DFAX: 29.4%

Pre-project loading: 9.8%

13. Facility: Greenwood-Marlton 138.0 kV Line

Contingency: Hell Creek-Framingham 138.0 kV

Loading: 205.8% (Rating: 214 MVA)

DFAX: 33.0%

Pre-project loading: 26.1%

14. Facility: Border-Marlton 138.0 kV

Contingency: Framingham-Hell Creek 138.0 kV

Loading: 181.7% (Rating: 228 MVA)

DFAX: 33.0%

Pre-project loading: 35.3%

15. Facility: Tomac-Wilmot 138.0 kV Line

Contingency: Centralf-Windhub 138.0 kV

Loading: 174.1% (Rating: 240 MVA)

DFAX: 19.6%

Pre-project loading: 51.7%

16. Facility: Wilmot-Seaway Sw Substation 138.0 kV

Contingency: Centralf-Windhub Substation 138.0 kV

Loading: 170.6% (Rating: 233 MVA)

DFAX: 19.6%

Pre-project loading: 44.5%

17. Facility: Border-Seaway Sw 138.0 kV

Contingency: Central-Windhub 138.0 kV

Loading: 169.7% (Rating: 218 MVA)

DFAX: 19.6%

Pre-project loading: 34.6%

18. Facility: Vincent-Crows Landing 138.0 kV

Contingency: Hell Creek-Windhub 138.0 kV

Loading: 167.2% (Rating: 185 MVA)

DFAX: 21.8%

Pre-project loading: 8.8%

19. Facility: Mout Laurel-University Wps 138.0 kV L

Contingency: Greenwood-Central 138.0 kV

Loading: 163.0% (Rating: 245 MVA)

DFAX: 29.4%

Pre-project loading: 16.9%

20. Facility: Philo-Corith 138.0 kV

Contingency: Oakland X-Mount Laurel 138.0 kV

Loading: 152.7% (Rating: 220 MVA)

DFAX: 14.1%

Pre-project loading: 56.6%

21. Facility: Vincent-Lumberton 138.0 kV

Contingency: Hell Creek-Windhub 138.0 kV L

Loading: 152.3% (Rating: 206 MVA)

DFAX: 21.4%

Pre-project loading: 3.5%

22. Facility: Oakland X-Mount Laurel Substation 138.0 kV

Contingency: Mount Laurel-Philo 138.0 kV

Loading: 150.2% (Rating: 262 MVA)

DFAX: 14.9%

Pre-project loading: 64.7%

23. Facility: Lolo-Corinth 138.0 kV

Contingency: La Crosse Substation-Oakland X 138.0 kV

Loading: 137.3% (Rating: 242 MVA)

DFAX: 7.3%

Pre-project loading: 91.9%

24. Facility: Oakland X Substation-La Crosse 138.0 kV

Contingency: Lolo-Corinth 138.0 kV

Loading: 137.0% (Rating: 233 MVA)

DFAX: 7.5%

Pre-project loading: 89.1%

25. Facility: Mount Laurel-Philo Substation 138.0 kV

Contingency: Oakland X-Mount Laurel 138.0 kV

Loading: 134.5% (Rating: 272 MVA)

DFAX: 14.1%

Pre-project loading: 57.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Windhub-Hell Crk 138.0 kV	Rebuild line with higher-capacity conductor	\$1,866,000	100% (\$1,866,000)
Hell Crk-Framingham 138.0 kV	Rebuild line with higher-capacity conductor	\$5,199,000	100% (\$5,199,000)
Gaffney-Framingham 138.0 kV	Rebuild line with higher-capacity conductor	\$485,000	100% (\$485,000)
Gaffney-Reno 138.0 kV	Rebuild line with higher-capacity conductor	\$1,232,000	100% (\$1,232,000)
Centralf-Windhub Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$2,397,000	100% (\$2,397,000)
Reno-Vincent 138.0 kV L	Rebuild line with higher-capacity conductor	\$1,409,000	100% (\$1,409,000)
Greenwood-Centralf 138.0 kV	Rebuild line with higher-capacity conductor	\$2,920,000	100% (\$2,920,000)
Reno-Jackson Square 138.0 kV	Rebuild line with higher-capacity conductor	\$7,809,000	100% (\$7,809,000)
Windhub-Dillion 138.0 kV	Rebuild line with higher-capacity conductor	\$1,462,000	100% (\$1,462,000)
Country Club 2 Tap Substation-Dillion 138.0 kV	Rebuild line with higher-capacity conductor	\$2,882,000	100% (\$2,882,000)
Country Club 2 Tap-Jackson Square 138.0 kV	Rebuild line with higher-capacity conductor	\$2,627,000	100% (\$2,627,000)
University Wps-Centralf 138.0 kV	Rebuild line with higher-capacity conductor	\$1,252,000	100% (\$1,252,000)
Greenwood-Marlton 138.0 kV L	Rebuild line with higher-capacity conductor	\$4,867,000	100% (\$4,867,000)
Border-Marlton 138.0 kV line	Rebuild line with higher-capacity conductor	\$1,153,000	100% (\$1,153,000)
Wilmot-Tomac 138.0 kV	Rebuild line with higher-capacity conductor	\$2,579,000	100% (\$2,579,000)
Wilmot-Seaway Sw 138.0 kV	Rebuild line with higher-capacity conductor	\$1,090,000	100% (\$1,090,000)
Border-Seaway Sw 138.0 kV	Rebuild line with higher-capacity conductor	\$2,216,000	100% (\$2,216,000)
Vincent-Crows Landing 138.0 kV line	Rebuild line with higher-capacity conductor	\$4,725,000	100% (\$4,725,000)
Mout Laurel-University Wps 138.0 kV L	Rebuild line with higher-capacity conductor	\$10,740,000	100% (\$10,740,000)
Corinth-Philo 138.0 kV	Rebuild line with higher-capacity conductor	\$1,213,000	100% (\$1,213,000)
Lumberton-Vincent 138.0 kV	Rebuild line with higher-capacity conductor	\$12,270,000	100% (\$12,270,000)
Mount Laurel-Oakland X 138.0 kV line	Rebuild line with higher-capacity conductor	\$3,693,000	100% (\$3,693,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Lolo-Corinth 138.0 kV	Rebuild line with higher-capacity conductor	\$1,437,000	100% (\$1,437,000)
La Crosse-Oakland X 138.0 kV L	Rebuild line with higher-capacity conductor	\$5,419,000	100% (\$5,419,000)
Mount Laurel-Philo Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$10,809,000	100% (\$10,809,000)
	Total allocated to project		\$93,751,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.